

SEMESTER END EXAMINATIONS - JANUARY 2023

Program	: B.E.- Information Science and Engineering	Semester	: V
Course Name	: Software Engineering	Max. Marks	: 100
Course Code	: IS53	Duration	: 3 Hrs

Instructions to the Candidates:

- Answer one full question from each unit.

UNIT - I

- Compare and contrast plan driven processes and agile processes with suitable use cases. CO1 (07)
 - What is the need for extreme programming? Draw and describe the extreme programming cycle. CO1 (06)
 - Identify the advantages and limitations of waterfall model with neat diagram. CO1 (07)
- Illustrate with necessary diagram, advantages and disadvantages of the process model that emphasizes risk analysis. CO1 (07)
 - Is it possible to combine software process models for a given problem statement? Justify with suitable example. CO1 (08)
 - State four different phases of Rational Unified Process (RUP) with neat diagram. CO1 (05)

UNIT - II

- Differentiate between user and system requirements. List out various user and system requirements for Hostel food wastage management system. CO2 (08)
 - Explain three types of Non-functional requirements with an example. CO2 (07)
 - Discuss Requirements change management system with block diagram. CO2 (05)
- Describe the structure of a requirements document based on an IEEE standard for requirement documents. CO2 (10)
 - Discuss different types of checks that should be carried during the requirement validation process. CO2 (05)
 - Write functional and non-functional requirement for Mental health care- Patient Monitoring System. CO2 (05)

UNIT - III

- What do you mean by a software architectural pattern? Describe with a diagram the organization of the Model-View-Controller pattern. CO3 (08)
 - Draw a state diagram with a minimum of five states for a weather station system and explain your answer. CO3 (12)

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| 6. | a) | Explain the essential elements of design patterns. | CO3 | (05) |
| | b) | What are the activities to be carried out in object oriented design process of a system? How objects are identified? | CO3 | (08) |
| | c) | Describe open source development, explain general models of open source licensing. | CO3 | (07) |

UNIT- IV

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| 7. | a) | Differentiate between software testing and inspections with neat diagram? | CO4 | (06) |
| | b) | Define partition testing? What is the purpose of Equivalence partitioning, Explain with neat diagram? | CO4 | (06) |
| | c) | What are the benefits of Test driven development? Explain with neat diagram. | CO4 | (08) |
| 8. | a) | What are the Stages in the acceptance testing process and describe any stage? | CO4 | (05) |
| | b) | Illustrate Legacy system management with suitable example. | CO4 | (08) |
| | c) | With diagram, explain the general model of the reengineering process. | CO4 | (07) |

UNIT - V

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| 9. | a) | Describe software Pricing, Explain the factors affecting the software pricing. | CO5 | (06) |
| | b) | With a neat diagram, Explain Project scheduling process in detail. | CO5 | (09) |
| | c) | List & differentiate between plan driven and agile planning. | CO5 | (05) |
| 10. | a) | Explain in detail CMMI process improvement framework. | CO5 | (10) |
| | b) | Describe software measurement and its metrics in detail with an example. | CO5 | (10) |
